

SBEmacs commands

How to type a command. C- is the **Ctrl** key and M- is the **Esc** key.

C-f Keep **Ctrl** down and press **f**.

M-f Press **Esc** and let go of it, then press **f**.
or keep **Alt** (**Option** on a Mac) down and press **f**.

C-x C-s Keep **Ctrl** down and press **x**, then **s**. Then let go of **Ctrl**.

C-x b Keep **Ctrl** down and press **x**. Let go of **Ctrl**, then press **b**.

C-M-f Press **Esc** and let go of it, then keep **Ctrl** down and press **f**.

Files

C-x C-f open a file

C-x C-s save

C-x C-w save with another name

C-x i insert a file

C-x C-c quit

Moving

C-f C-b forward, back one char

C-n C-p next, previous line

M-f M-b forward, back one word

C-a C-e beginning, end of the line

C-v M-v next, previous screen

M-< M-> beginning, end of the file

M-g go to a line number

Editing

DEL delete char before the cursor

C-d delete char under the cursor

C-k kill to end of line

C-/ undo; again: undo more

C-M-_ redo

TAB indent the line

M-q refill the paragraph

M-; comment line or region

Region (shaded)

C-Spc set mark

mouse drag to select

C-w kill region

M-w copy the region

C-y yank (paste)

M-y after C-y: an earlier kill

C-g cancel command

Searching

C-s search forward

C-r search backward

M-% replace

Buffers and windows

C-x b choose a buffer from a filtered, completing list

C-x C-b buffer menu

C-x 2 split the window

C-x o other window (or click it)

C-x 0 close this window

C-x 1 close other window

Help, Lisp and AI

C-h help with keybinding

M-x run a command by name

M-: eval a Lisp

C-x C-e eval Lisp before cursor

C-c r call Claude

C-c y insert Claude's code

C-c g call Codex/ChatGPT

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Chapter 1

The commands in detail

This chapter explains the commands of the first page in more detail. The next chapter shows how to install SBEmacs.

In the following cheat sheet, **Spc** denotes the space bar, **C-** is the Ctrl prefix, **M-** represents the Alt prefix and κ can be any key.

Help. If you forget a command, press **C-h** or F1. SBEmacs shows all its key-bindings on a page of their own, one key per line. The arrows, **C-n** and **C-p** scroll the page; any other key takes you back to the file you are editing.

The status line is a menu. At the bottom of each window, the status line shows the name of the file and the line where the cursor is. An asterisk after the file name means that the file has been modified, but not saved. Then come buttons that the mouse can click:

```
fib.lisp*, L. 5 == SBEmacs: [HELP] [SAVE] [OPEN] [COPY] ...
```

A button does what its key does, so Nia can work before she learns the keys, and learn them from the buttons:

[HELP]	C-h	every key on one page
[SAVE]	C-x C-s	save the file
[OPEN]	C-x C-f	open a file
[COPY]	M-w	copy the selection
[PASTE]	C-y	paste
[UNDO]	C-/	undo
[AI HELP]		how to ask Claude or Codex
[BUFFERS]	C-x b	switch to another open file
[QUIT]	C-x C-c	leave SBEmacs

The name of the file is a button too. A name longer than 18 characters is shortened at the front, as in `...src/fib.lisp`. A click on the name opens a window with the full name of the file, its number of lines, words, characters and bytes, the line of the cursor, the language, and whether and when it was saved. Any key closes that window. To copy, Nia drags the mouse over the text and clicks [COPY]; she then clicks where the copy should go, and clicks [PASTE]. A button lights up while she presses it, and acts when she lets it go; if she changes her mind, she moves the mouse off the button before letting go. If Nia leaves SBEmacs with [QUIT] while a file is not saved, SBEmacs asks first.

The whole status line is written in Lisp, in `lisp/extensions/menu.lisp`. The buttons are a list, `*mode-line-menu*`, which Nia can change in her init file:

```
(setf *mode-line-menu*
      (append *mode-line-menu* '(("KILL" "C-x k"))))
```

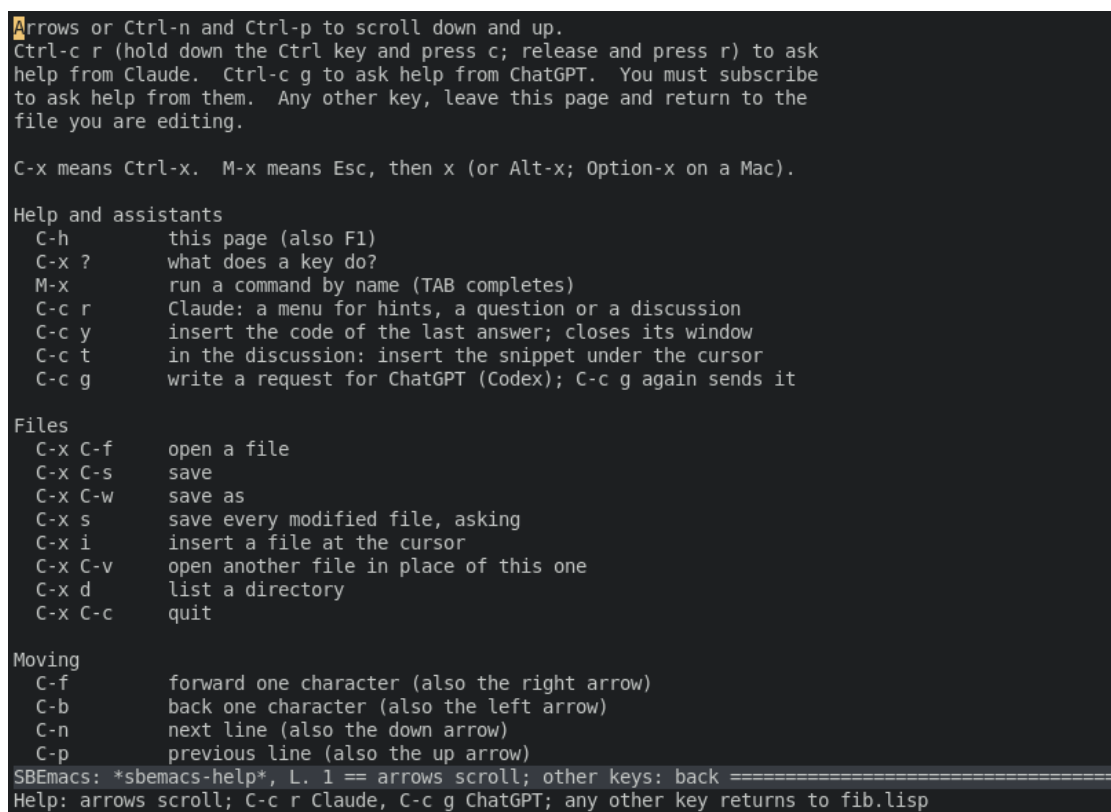


Figure 1.1: The help page, C-h

Basic commands. C- κ – Keep **Ctrl** down and press κ

C-s then type some text, and press C-s again – search a text.
 C-r then type some text, and press C-r again – reverse search.
 C-k – kill a line.
 C-h – help (the backspace key deletes backward).
 C-d – delete char forward.
 C-Spc then move the cursor – select a region.
 M-w – save selected region into the kill ring.
 C-w – kill region.
 C-y – insert killed or saved region.
 C-g – cancel minibuffer reading, or unselect the region.
 C-a – go to beginning of line.
 C-e – go to end of line.
 C-b – move backward one character.
 C-f – move forward one character.
 C-n – move cursor to the next line.
 C-p – move cursor to the previous line.
 C-l – put the line of the cursor in the middle of the window.
 C-/ – undo. C-u also undoes.
 C-v – forward page.
INS – toggle overwrite mode.
← **→** **↑** **↓** – arrows move cursor.

Control-x commands. C-x C- κ – Keep **Ctrl** down and press **x**, then κ

C-x C-f – open a file into a new buffer.
 C-x C-w – write file with new name.
 C-x C-s – save file.
 C-x s – save every modified file, asking first.
 C-x C-c – exit SBEmacs.
 C-x i – insert file at the cursor.
 C-x ? – describes a key stroke.
 C-x @ – shell command. After C-x @, if one types **ls** at the prompt, the computer will list the current folder. It is possible to use any other shell command just like the **ls** example.

Window commands. One can have more than one window on the screen. Below, you will find commands that cope with this situation.

- C-x κ – Keep **Ctrl** down and press **x**; let go of **Ctrl**, then press κ

C-x b – choose a buffer. Type to filter the visible list, **Tab** completes the common prefix, the arrow keys select an entry, **Enter** opens it, and **C-g** cancels. The previous buffer is selected initially, so repeated use quickly alternates between two buffers.

C-x n – next buffer.

C-x C-n – next buffer. Same as above.

C-x C-b – list buffers.

C-x k – kill current buffer.

C-x = – cursor position.

C-x 2 – split window into cursor window and other window.

C-x o – jump to the other window.

C-x 1 – close the other window.

C-x 0 – close the cursor window.

C-x ^ – make the cursor window taller.

- **Esc κ** – Press the **Esc** key and let go of it, then press the **κ** key.

Esc > – go to the end of buffer.

Esc < – go to the beginning of buffer.

Esc f – word forward.

Esc b – word backward.

- **M- κ** – Keep the **Alt** key down and press the **κ** key. On the Macintosh, the **Alt** key is called **Option**; in the Terminal application, you must tick *Use Option as Meta key* in Settings, Profiles, Keyboard. In iTerm2, open Settings, Profiles, Keys and set *Left Option key* to *Esc+*. The right Option key may remain *Normal* for entering Mac symbols.

M-b – move backward one word.

M-f – move forward one word.

M-g – go to the line given in the minibuffer.

M-n – activate the next buffer.

M-r – query replace. **M-%** does the same.

The mouse. You can click on a window to move the cursor there; a click on another window selects that window. If you keep the button down and drag the mouse, you select a region. This works in the graphical window and in the text terminal. In a text terminal, SBEmacs takes the mouse from the terminal; if you want the terminal's own selection, hold down the **Shift** key (the **Option** key in the Macintosh Terminal) while you drag, or start the editor with the `--no-mouse` option.

The region. The region is the text between the mark, which you set with `C-Spc`, and the cursor. Like in Emacs, SBEmacs shades the region while it is active. The shade goes away when you change the text, when you copy the region with `M-w`, or when you press `C-g`.

Query replace. If you press `[Esc]r`, the computer enters in the query replace mode. First, SBEmacs prompts for the text snippet `S` that you want to replace. Then it prompts for the replacement `R`. When you type the `R` text and press the `[Enter]` key, the cursor jumps to the first occurrence of `S`, and SBEmacs asks whether you want to replace it. If the answer is `y`, SBEmacs will replace `S` with `R` and jump to the next occurrence. If the answer is `n`, SBEmacs will jump to the next occurrence of `S` without performing the replacement.

Go to line. When you try to compile code containing errors, the compiler usually reports the line number where the error occurred. If you press `[Esc]g`, SBEmacs prompts for a line number. As soon as you type the number and press the `[Enter]` key, the cursor jumps to the line where the error occurred.

In order to know the position of the cursor, press the `C-x =` command, and SBEmacs gives information concerning the character under the cursor, the corresponding code, the line, and the character position in the text.

Markdown. SBEmacs colours Markdown files (`.md`): headings, **`**strong**`** and **`*emphasis*`**, code between back quotes, links, quotations and lists. A block of code that names its language, like ````lisp`, is coloured as that language. In a list, `[Enter]` starts the next item, `-` or `3.` ; `[Enter]` on an empty item ends the list. `[Tab]` nests an item under the item above, and `M-q` refills a paragraph or a list item.

Undo. The `C-/` command undoes the last command. If you press it again, it undoes the command before, and so on. Typed text is undone a word at a time, and a command that changed many lines, like `M-q` (fill paragraph), is undone at once. If you undo too much, `C-M-_` (or `M-_`) redoes what undo took back. When you undo all the changes made since the file was saved, the asterisk disappears from the status line.

Nia Vardalos. In order to get acquainted with editing and scripting, let us accompany the Canadian programmer Nia Vardalos, while she explores SBEmacs.

When text is needed for carrying out a command, it is read from the minibuffer, a one line window at the bottom of the screen. For instance, if Nia presses `C-s` for finding a text, the text is read from the minibuffer. After typing the text that

she wants to localize, Nia presses **C-s** again to start the search. For abandoning an ongoing command, she may press **C-g**.

To transport a region from one place to another, Nia presses **C-Spc** to start the selection process and move the cursor to select the region. Then she presses **C-w** to kill her selection. Finally, she moves the cursor to the insertion place, and presses the **C-y** shortcut. Of course, she can select the region with the mouse instead.

To copy a region, Nia presses **C-Spc** and moves the cursor to select the region. Then she presses **M-w** to save the selection into the kill ring. Finally, she takes the cursor to the destination where the copy is to be inserted and issues the **C-y** command. If she wants an earlier kill, she presses **M-y** right after **C-y**, and SBEmacs replaces the yanked text with the kill before it.

The kill ring and the system clipboard work together, as in Emacs. Whatever Nia kills or copies with **C-w**, **M-w** or **C-k** also goes to the clipboard, so she can paste it into any other program. And when she copies something in another program, say a paragraph in her browser, **C-y** inserts it, and it joins the kill ring; **M-y** then goes on to her older kills. In a terminal, SBEmacs reaches the clipboard through **pbcopy** and **pbpaste** on the Macintosh, and through **wl-copy** or **xclip** on Linux (install one of them); in a window, it needs nothing.

When Nia selects text by dragging the mouse, SBEmacs copies it to the clipboard at once: **Cmd-V** pastes it in her browser, and she never needs **Cmd-C**. In fact, in a terminal **Cmd-C** finds nothing to copy, because SBEmacs takes the mouse for itself (that is how the buttons of the status line work); iTerm2 then complains that mouse reporting has prevented making a selection. The text is already on the clipboard. If Nia wants the terminal's own selection, she keeps **Option** down while she drags.

Nia noticed that there are two equivalent ways to issue an **M-κ** command. She can press and release the **Esc** key, then strike the κ key. Alternatively, she can keep the **Alt** key down and press the κ key.

Calculations with Lisp. SBEmacs offers a Lisp dialect to perform calculations and text processing. Let us assume that you want to find how many students graduate from medical schools in California. You press **Esc** **:** and SBEmacs will prompt you for a Lisp expression.

```
> (+ 190 180 170 160 120 100 100 90) Enter
```

In the above expression, the first thing after the open parenthesis is the **+** sum operation identifier. After the name of the operation, there is a list of the arguments to be added together. As soon as Nia presses the **Enter** key, SBEmacs shows the result of the addition:

```
1110
```


The rule that worked for the $(+ x_1 x_2 \dots)$ sum operation also works for the other arithmetic function too.

- Multiplication is performed by the $*$ operation.

```

Esc :
> (* 1 2 3 4 5) Enter
120

```

- Division is obtained through the $/$ operation.

```

Esc :
> (/ 1 2 4) Enter
1/8

```

The above operation performed the chain division $1/2/4$. Common Lisp keeps the exact fraction $1/8$. If you want a decimal number, write one of the arguments as a decimal: $(/ 1.0 2 4)$ gives 0.125 .

- Subtraction has the syntax shown below.

```

Esc :
> (- 1 2 4) Enter
-5

```

- Mixed operations. One can nest subexpressions within an outer expression. For instance, to get the average student number graduating from medical school in California, one can calculate the expression:

```

Esc :
> (/ (+ 190 180 170 160 120 100 100 90) 8.0) Enter
138.75

```

In FemtoEmacs, the predecessor of SBEmacs, the Lisp prompt was `Esc ;`. In SBEmacs, as in GNU Emacs, `Esc ;` inserts a comment, and the Lisp prompt is `Esc :`.

Command execution. When you press shortcut keys, SBEmacs calls a command written in C or in Lisp. You can get a complete list of all SBEmacs key-bindings by pressing `C-h`.

There are commands that are not associated to shortcut keys. For instance, the `undo-status` command tells how many steps of undo the buffer keeps. Press `Esc x` and type `undo-status` at the minibuffer prompt. The `Tab` key completes the name of the command. Below you will find a fairly complete list of SBEmacs commands.

Moving.

C-f	forward one character (also the right arrow)
C-b	back one character (also the left arrow)
C-n	next line (also the down arrow)
C-p	previous line (also the up arrow)
M-f	forward one word
M-b	back one word
C-a	beginning of the line
C-e	end of the line
M-m	first non-blank character of the line
M-a	back to the start of the sentence
M-e	forward to the end of the sentence
M-{	back one paragraph
M-}	forward one paragraph
C-v	next screen
M-v	previous screen
M-<	beginning of the file
M->	end of the file
M-g	go to a line number
C-l	put the cursor's line in the middle of the window
C-M-f	forward over an expression (Lisp: an s-expression)
C-M-b	back over an expression
C-M-u	out to the enclosing parenthesis
C-M-d	into the next parenthesis
C-M-a	beginning of the definition (defun, function)
C-M-e	end of the definition

Selecting. The region is shaded.

C-Spc	set the mark: the region runs from it to the cursor
mouse	click to move the cursor or choose a window; drag to select
C-x h	select the whole file
M-h	select the paragraph
M-@	select to the end of the word
C-M-Spc	select to the end of the expression
C-x C-x	swap the cursor and the mark
C-g	unselect; cancel a command

Killing (cutting) and yanking (pasting).

C-w	kill the region
M-w	copy the region

C-y	yank the last kill
M-y	right after C-y: replace it with the kill before
C-k	kill to the end of the line (again: joins the kills)
M-d	kill the next word
M-DEL	kill the previous word
M-k	kill to the end of the sentence
C-M-k	kill the next expression
M-z	kill up to and including a character
C-d	delete the character under the cursor
DEL	delete the character before the cursor (Backspace)
C-c i	choose a kill to insert
C-c k	browse the kill ring

Changing text.

C-/	undo the last command; again: the one before
C-M-_	redo what undo took back (also M-_)
C-t	swap the two characters at the cursor
M-t	swap two words
C-x C-t	swap this line with the one above
M-u	upper-case the next word
M-l	lower-case the next word
M-c	capitalize the next word
C-x C-u	upper-case the region
C-x C-l	lower-case the region
C-o	open a line after the cursor
C-j	new line, indented
TAB	indent the line
C-x TAB	indent the region (also C-M-\)
M-^	join this line to the previous one
M-\	delete the spaces around the cursor
M-Spc	leave just one space
C-x C-o	delete blank lines
M-q	refill the paragraph or comment
M-;	comment: add one, or (un)comment the region
C-q	insert the next key as it is (C-q TAB: a tab)
INS	overwrite mode on or off

Searching, buffers, windows and macros.

C-s	search forward (C-s again: next match)
C-r	search backward
M-%	replace, asking each time (also M-r)
C-x C-g	grep: search files

C-x !	next grep match
C-x b	choose a buffer from a filtered, completing list
C-x C-b	buffer menu
C-x 2	split the window
C-x o	go to the other window (or click it)
C-x 0	close this window
C-x 1	close the other windows
C-x (	start recording keys
C-x)	stop recording
C-x e	replay them; then e replays again

Lisp and assistants.

C-x C-e	evaluate the expression before the cursor
C-M-x	evaluate the top-level form around the cursor
M-:	evaluate an expression typed at the bottom
M-]	evaluate the block and insert the value
M-!	a window for a shell command; C-c s runs it
C-x C-r	reload the Lisp scripts (no rebuild needed)
C-c r	Claude: hints, a question or a discussion
C-c y	insert the code of the last answer
C-c t	in the discussion: insert the snippet under the cursor
C-c g	write a request for ChatGPT (Codex)

Shell commands. M-! opens a window below the file, where Nia writes a command for the operating system, say `ls -l` or `sbcl --script fib.lisp`. A long command may take several lines: `Enter` breaks the line, and SBEmacs joins the lines again as the system expects (with `\` at the end of each line on Linux and the Macintosh). C-c s runs the command in the folder of her file. The window then shows the command and, below it, what the command printed; if it failed, its exit status too. C-x 1 closes the window. Its status line has two buttons, so Nia can click instead of typing:

```
SBEmacs: *shell-command*, L. 1 == [SEND - C-c s] [CLOSE - C-x 1]
```

Chapter 2

Installation

A text editor, like Emacs or SBEmacs, has three components:

1. A frontend that inserts characters into a buffer. A buffer is a memory region with an image of a text file. One can see the buffer as a representation of a scratch pad. As such, the frontend provides tools to insert keystrokes at the point indicated by a movable cursor.
2. Editing operations that save buffers as files, delete and insert text, move the cursor around, copy items from one place to another, etc.
3. A scripting language for customizing the editor.

There is a large choice of frontends. Some frontends work with a raster graphics image, which is a dot matrix data structure that represents a grid of pixels. There are also frontends that are specialized in showing letters and other characters via an appropriate display media. The latter sort of frontend is called text-based user interfaces, while the former is called graphical user interface, or GUI for short.

Text-based user interfaces are more comfortable on the eyes, since they provide sharper and crisp alphabetical letters. On the other hand, GUI allows for font customization. SBEmacs offers both. In a text terminal, it uses ncurses, which is a text-based user interface. In a window of its own, it uses SDL2, a library that draws on a grid of pixels, and SDL2_ttf, which renders TrueType fonts. The two frontends are compiled into two separate libraries, so a person who only works in the text terminal never needs SDL2.

The scripting language is Common Lisp, as implemented by SBCL. In fact, SBCL is the program that you run: it loads the small editor core, written in C, and then the Lisp scripts.

In this chapter, you will install all these pieces. I have to say this again: An installation task has so many details that, unless you are an engineer yourself, ask

for help from a computer expert. If you never used a text terminal, read Chapter 3 first.

2.1 Cloning the GitHub repository

The sources of SBEmacs are kept on GitHub, a site that stores programs together with the history of all their changes. Such a collection is called a repository. To get a copy of the repository on your machine, you need the `git` program. In order to check whether you have it, type:

```
~$ git --version
git version 2.39.5
```

If the computer says that it does not know `git`, install it. On the Macintosh, the command below installs `git` together with the C compiler and `make`, which you will need anyway:

```
~$ xcode-select --install
```

On Debian or Ubuntu Linux, use `apt-get`:

```
~$ sudo apt-get install git build-essential
```

Now, clone the repository, i.e., make a local copy of it. Let us assume that Nia keeps her sources in a `csand` folder:

```
~$ mkdir -p csand
~$ cd csand
~/csand$ git clone https://github.com/FemtoEmacs/Femto-Emacs.git
↪ sbemacs
~/csand$ cd sbemacs
~/csand/sbemacs$
```

The last argument of `git clone`, `sbemacs`, is the name of the folder that receives the copy. If you omit it, `git` names the folder after the repository, `Femto-Emacs`.

Later on, when the authors improve the editor, you do not need to clone the repository again. Enter the folder and ask `git` to bring the changes:

```
~/csand/sbemacs$ git pull
```

2.2 Branches

A repository can keep more than one version of a program at the same time. Each version is called a branch. The SBEmacs repository (still called **Femto-Emacs** on GitHub) has three branches:

- **master** – FemtoEmacs, the predecessor of SBEmacs (2016), with femtolisp.
- **sbcl** – SBEmacs, the editor described in this text, for Linux, Macintosh and Windows.
- **windows** – SBEmacs with an installer for Windows, so that Windows users do not need a C compiler.

The `git branch -a` command lists the branches; an asterisk marks the one whose files you have in the folder. To switch to another branch, use `git switch`:

```
~/csand/sbemacs$ git branch -a
* master
  remotes/origin/master
  remotes/origin/sbcl
  remotes/origin/windows
~/csand/sbemacs$ git switch sbcl
Switched to a new branch 'sbcl'
~/csand/sbemacs$ git switch windows
Switched to a new branch 'windows'
~/csand/sbemacs$ git switch sbcl
Switched to branch 'sbcl'
```

When you switch, git replaces the files of the folder with the files of the chosen branch. Therefore, if you have changed a file, git refuses to switch until you save your work with `git commit`, or put it aside with `git stash`. Old versions of git do not know `git switch`; use `git checkout sbcl` instead.

For the rest of this chapter, Nia stays in the **sbcl** branch. If you use Windows and only want to run the editor, switch to the **windows** branch and follow its `README.md`: it explains how to install SBCL and run the installer, which needs neither a C compiler nor administrator privileges.

2.3 Installing SBCL

SBCL, Steel Bank Common Lisp, is a free compiler for the Common Lisp language. Users of SBEmacs are supposed to install it, because SBEmacs is a program that SBCL runs. Any recent version works.

MAC OS-X. The easiest way is Homebrew, a package manager for the Macintosh (see section 3.1). If you do not have Homebrew yet, the instructions are at <https://brew.sh>. Then:

```
~$ brew install sbcl
```

Linux. Use the package manager of your distribution:

```
~$ sudo apt-get install sbcl      # Debian, Ubuntu
~$ sudo dnf install sbcl         # Fedora
```

Windows. Download the installer of the most recent 64-bit version from <https://www.sbcl.org/platform-table.html>, double-click it and follow the instructions.

Checking. Whatever your system, check that the terminal finds SBCL:

```
~$ sbcl --version
SBCL 2.5.9
```

If you type `sbcl` alone, you get the Lisp prompt, where you can try the arithmetic of Chapter 1. The `(exit)` expression takes you back to the shell.

```
~$ sbcl
* (+ 190 180 170 160 120 100 100 90)
1110
* (exit)
~$
```

2.4 ncurses

Computers need a tool to make communication with users possible. That tool is called User Interface, or UI for short. A text terminal offers sharper and crisp characters through the use of a specialized User Interface called ncurses. SBEmacs uses ncurses when it runs in a text terminal, and needs the version that handles characters with diacritics, so you can write words like *café* and *façade*. This version is called ncursesw, where the `w` stands for wide characters.

In 2016, one had to download ncurses and build it, because the operating systems came with old versions. Today, this is seldom necessary.

MAC OS-X. The Macintosh comes with ncurses, and SBEmacs uses it. The Xcode command line tools of section 2.1 bring the files that the C compiler needs. To check the version:

```
~$ tic -V
ncurses 6.0.20150808
~$ ls $(xcrun --show-sdk-path)/usr/include/ncurses.h
/Library/Developer/CommandLineTools/SDKs/MacOSX.sdk/usr/include/ncurses.h
↪ rses.h
```

If `xcrun` complains, run `xcode-select --install` again.

Linux. The ncurses library is certainly installed, since many programs use it, but the files that the C compiler needs come in a separate package. Check whether the compiler can find ncursesw:

```
~$ pkg-config --modversion ncursesw
6.4.20240113
```

If the computer answers that the package was not found, install it:

```
~$ sudo apt-get install libncurses-dev pkg-config # Debian, Ubuntu
~$ sudo dnf install ncurses-devel pkgconf # Fedora
```

Windows. If you build SBEmacs from the sources, you need MSYS2 (<https://www.msys2.org>), a collection of Unix tools for Windows. In the *MSYS2 MINGW64* shell, install the compiler, make, ncurses and pkg-config:

```
$ pacman -S make mingw-w64-x86_64-gcc mingw-w64-x86_64-ncurses \
mingw-w64-x86_64-pkgconf
```

2.5 *SDL2 and SDL2_ttf*

What they are. A text terminal shows only characters, one per cell of a grid. A window of its own lets SBEmacs choose fonts and colours freely, but the program must then draw every letter itself, pixel by pixel, and talk to the operating system to open the window and to read the keyboard and the mouse. Each operating system does these things in its own way: Cocoa on the Macintosh, X11 or Wayland on Linux, Win32 on Windows. Instead of writing the editor three times, SBEmacs uses SDL2.

SDL2, the Simple DirectMedia Layer, is a library that offers one set of functions for all these systems: open a window, draw rectangles and images, receive key

presses and mouse clicks. Many games are built on it. SDL2 does not know how to draw letters, though. That is the job of `SDL2_ttf`, a companion library that reads TrueType fonts (the `.ttf` and `.ttc` files of your system) and turns each letter into a small image that SDL2 can paint.

Remember that a library is a file of compiled C functions that other programs use. To build a program on top of a library, the C compiler needs two things: the header files (`SDL.h`, `SDL_ttf.h`), which describe the functions, and the library itself (`libSDL2.dylib`, `libSDL2.so` or `SDL2.dll`). On Linux, the headers come in a separate package, whose name ends in `-dev` or `-devel`.

SDL2 and `SDL2_ttf` are optional. If you only use SBEmacs in a text terminal, you can skip this section.

pkg-config. How does `make` find the headers and the libraries? It asks `pkg-config`, a small program that knows where each installed library lives. For instance:

```
~$ pkg-config --modversion sdl2 SDL2_ttf
2.30.0
2.22.0
~$ pkg-config --cflags --libs sdl2 SDL2_ttf
-I/opt/homebrew/include/SDL2 -L/opt/homebrew/lib -lSDL2_ttf -lSDL2
```

The first command prints the versions, which proves that both libraries are installed and that `pkg-config` finds them. The second prints the options that `make` hands to the C compiler. If the first command answers *Package sdl2 was not found*, install the libraries as shown below. If it answers *command not found*, `pkg-config` itself is missing; install it too.

MAC OS-X. Install Homebrew first, as explained in section 3.1. Then:

```
~$ brew install sdl2 sdl2_ttf pkg-config
```

Homebrew installs everything under `/opt/homebrew` (or `/usr/local` on older Intel Macintosh computers), and tells `pkg-config` where. You do not need `sudo`.

Linux. Install the development packages, which bring the headers together with the libraries:

```
~$ sudo apt-get install libsdl2-dev libsdl2-ttf-dev pkg-config #
↪ Debian, Ubuntu
~$ sudo dnf install SDL2-devel SDL2_ttf-devel pkgconf #
↪ Fedora
~$ sudo pacman -S sdl2 sdl2_ttf pkgconf #
↪ Arch
```

SBEmacs needs a font. Most distributions come with the DejaVu fonts; if letters are missing in the window, install them (`sudo apt-get install fonts-dejavu`).

Windows. If you install SBEmacs with the installer of the `windows` branch, SDL2 and `SDL2_ttf` come with it, and you have nothing to do. If you build SBEmacs yourself, install them in the `MSYS2 MINGW64` shell:

```
$ pacman -S mingw-w64-x86_64-SDL2 mingw-w64-x86_64-SDL2_ttf \
    mingw-w64-x86_64-pkgconf
```

The files `SDL2.dll` and `SDL2_ttf.dll` must then be where Windows finds them: in the folder of `sbemacs.exe`, or in the `MSYS2 mingw64\bin` folder if it is in the `PATH`.

Build again. The graphical frontend is compiled only when `make` finds SDL2. If you ran `make` before installing SDL2, it printed this note, and built the terminal version only:

```
note: SDL2/SDL2_ttf not found, built the terminal version only
```

After installing SDL2, build and install again:

```
~/csand/sbemacs$ make
~/csand/sbemacs$ sudo make install
```

To make sure that the graphical frontend exists, look for its library next to the program:

```
~/csand/sbemacs$ ls libsbemacs-*
libsbemacs-gui.dylib  libsbemacs-term.dylib
```

On Linux the names end in `.so`; on Windows, in `.dll`.

Try it.

```
~$ sbemacs --gui notes.md
```

If a window opens, everything is in place. If SBEmacs says *cannot open a window here, using the terminal*, either the graphical library was not built (go back to *Build again*) or there is no screen to open a window on: for instance, you are logged in a remote machine through `ssh`. In that case the terminal version is the right one.

Fonts and colours. SBEmacs looks for a monospaced font among the usual ones of each system (Menlo and Monaco on the Macintosh, DejaVu Sans Mono and Liberation Mono on Linux, Consolas on Windows). To choose another one, give the path of its file in `~/.sbemacs/init.lisp`:

```
(setf *gui-font* "/Library/Fonts/JetBrainsMono-Regular.ttf")
(setf *gui-font-size* 16)
(setf *gui-theme* :light)           ; or :dark, the default
```

In the window, `C-+` and `C--` (or `Cmd` `[+]` and `Cmd` `[-]` on the Macintosh) make the letters larger or smaller.

2.6 make and sudo make install

The editor folder contains many files in the C computer language, and many in Lisp. In order to become usable, these files must be compiled into a set of libraries containing instructions written in a language that the computer understands. You do not need any knowledge of C to perform the translation to machine language. All instructions in the batch job for performing the compilation are written in Makefile. Enter the folder and type `make`:

```
~/csand/sbemacs$ git switch sbcl
~/csand/sbemacs$ make
~/csand/sbemacs$ make test
112/112 tests passed
```

The `make` command produces three files:

- `libsbemacs-term` – the editor core with the ncurses frontend.
- `libsbemacs-gui` – the editor core with the SDL2 frontend, if SDL2 is installed.
- `sbemacs` – the program itself, which SBCL saved together with the compiled Lisp scripts.

On the Macintosh, the libraries have the `.dylib` extension; on Linux, `.so`; on Windows, `.dll`. The `make test` command is optional: it checks that the Lisp side works.

You can already try the editor from inside the folder. Remember that the dot indicates the current directory. For instance, the `./sbemacs` command means: execute the `sbemacs` file present in the current directory.

```
~/csand/sbemacs$ ./sbemacs samples/fib.lisp
```

Installation is achieved by copying the compiled files to `/usr/local/`:

```
~/csand/sbemacs$ sudo make install
```

The installation requires that the system copy files to the `/usr/local/` folder. However, you must give special permission for this operation to be performed, otherwise the possibility remains open for a virus to make similar copies in critical folders. The `sudo` command informs the machine that you are a superuser. To avoid being tricked by a virus or an unauthorized user, the machine asks for the password, and if you type it correctly, the system installs SBEmacs: the program goes to `/usr/local/lib/sbemacs/`, and the commands `sbemacs` and `sbemacs-gui` to `/usr/local/bin/`.

Only `sudo make install` needs `sudo`. Never type `sudo make`: the files that a superuser creates belong to the superuser, and the next `make` cannot overwrite them. If you prefer not to use `sudo` at all, install SBEmacs in your home folder:

```
~/csand/sbemacs$ make install PREFIX=$HOME/.local
```

and make sure that `~/local/bin` is in your search path (in the `~/zshrc` or `~/bashrc` file):

```
export PATH="$HOME/.local/bin:$PATH"
```

Every time you bring a new version with `git pull`, repeat `make` and `sudo make install`. However, if you only change a Lisp script, you do not need to build anything: see section 2.10.

2.7 Running SBEmacs

In a text terminal, type `sbemacs` followed by the name of the file you want to edit:

```
~$ sbemacs fib.lisp
```

To open the editor in a window of its own, add the `--gui` option, or use the `sbemacs-gui` command:

```
~$ sbemacs --gui fib.lisp
~$ sbemacs-gui fib.lisp
```

If a window cannot be opened, for instance when you are logged in a remote machine, SBEmacs says so and works in the terminal. Other options:

```
;; Fibonacci, the naive way
(defun fib (n)
  (if (< n 2)
      n
      (+ (fib (- n 1)) (fib (- n 2)))))

(defun sq (x) (* x x))
```

SBEmacs: Ctrl-h for help == fib.lisp, L. 5 == Ctrl c r calls Claude; Ctrl c g calls GPT =====
 Ask Claude

h Hints: Claude reads the code at the cursor (and the selected region) and suggests fixes and next steps.
 q Question: write it in a window of its own, as many lines as you like; C-c r sends it.
 d Discussion: a conversation in a window that stays open; C-c t there inserts the snippet under the cursor into your file.
 s Set-up: is Claude installed and logged in?

Any other key closes this menu. Claude only reads: C-c y inserts the code it proposes, after you confirm.

SBEmacs: *assistant*, L. 1 -- h hints; q ask; d discuss; s set-up -----
 Claude: h hints, q question, d discussion, s set-up; any other key closes the menu

Figure 2.1: `sbemacs -gui fib.lisp`: a region selected with the mouse, and the menu of `C-c r`

- q – do not load your configuration file, `~/.sbemacs/init.lisp`.
- no-mouse – leave the mouse to the terminal.
- version – print the version and exit.

In the window, the usual shortcuts of the Macintosh work too: `Cmd C` copies, `Cmd X` cuts, `Cmd V` pastes, `Cmd Z` undoes and `Cmd Q` quits. `C-+` and `C--` make the font larger or smaller. In `~/.sbemacs/init.lisp`, you can choose the font, its size and a light or dark theme:

```
(setf *gui-font-size* 16)
(setf *gui-theme* :light) ; or :dark
(setf *gui-font* "/Library/Fonts/Menlo.ttc")
```

2.8 Installing Claude and Codex

SBEmacs can ask Claude, an artificial intelligence made by Anthropic, for help with the text you are editing. The editor does not talk to Claude directly: it calls

Claude Code, a program that you install and log in once. You must subscribe to Claude: Claude Code requires a Pro, Max, Team or Enterprise plan, or an Anthropic Console account; the free plan does not include it.

MAC OS-X and Linux.

```
~$ curl -fsSL https://claude.ai/install.sh | bash
```

On the Macintosh, Homebrew works too:

```
~$ brew install --cask claude-code
```

Windows. In PowerShell:

```
PS C:\Users\nia> irm https://claude.ai/install.ps1 | iex
```

Logging in. Open a new terminal, check the installation, and run `claude` once. It opens the browser, where you log in with your Claude account.

```
~$ claude --version
~$ claude
```

After logging in, type `/exit` to leave Claude Code. You will not need to log in again.

API key. If you have an Anthropic API key instead of a subscription, SBEmacs can use it: write the key in the first line of the `~/.sbemacs/anthropic-api-key` file, or put it in the `ANTHROPIC_API_KEY` environment variable. Never put the key in a file that you share or send to GitHub.

Codex and ChatGPT. The `C-c g` command asks Codex, OpenAI's programming assistant, using an eligible ChatGPT subscription. Install the current Codex program, for instance with `npm install -g @openai/codex@latest`, run `codex` once and choose *Sign in with ChatGPT*. On Windows, PowerShell can install it without npm:

```
PS C:\Users\nia> irm https://chatgpt.com/codex/install.ps1 | iex
PS C:\Users\nia> codex
```

On a Macintosh, SBEmacs also looks for the copy of Codex inside the ChatGPT application. If that copy needs authentication, the same one-time terminal sign-in completes the set-up. The editor never asks for an API key.

Checking. Inside SBEmacs, press `C-c r s` to check Claude or `C-c g s` to check Codex. The editor tells whether it found the corresponding program, where it found it, and how to fix what is missing.

2.9 Using Claude and Codex

The keybinding that calls Claude was inspired by Isaac Asimov’s robot stories, where robots have names like R. Daneel Olivaw and R. Giskard: `C-c r` is R. for robot. The assistant also obeys Asimov’s Three Laws of Robotics:

1. **First Law.** “A robot may not injure a human being or, through inaction, allow a human being to come to harm.” Claude never changes your text by itself. It sees only what `C-c r` sends, and it runs with all its tools disabled, so it cannot touch your files.
2. **Second Law.** “A robot must obey the orders given it by human beings except where such orders would conflict with the First Law.” Claude types into your buffer only when you order it with `C-c y` or `C-c t`, it asks before doing so, and `C-/` takes the insertion back.
3. **Third Law.** “A robot must protect its own existence as long as such protection does not conflict with the First or Second Law.” A missing program, a network error or a timeout end in a message, never in a crash or a lost buffer.

When Nia presses `C-c r`, a small menu opens below her file:

```
h  Hints: Claude reads the code at the cursor (and the selected
   region) and suggests fixes and next steps.
q  Question: write it in a window of its own, as many lines as
   you like; C-c r sends it.
d  Discussion: a conversation in a window that stays open; C-c t
   there inserts the snippet under the cursor into your file.
s  Set-up: is Claude installed and logged in?
```

Any other key closes the menu. In every window that belongs to Claude, the status line tells which keys work there.

Hints. If Nia presses `h`, Claude reads the text around the cursor and, if she selected a region, the region too. The answer appears in a window below the file. While Claude thinks, the editor waits; the message line says so.

Questions. If Nia presses `q`, a window called `*claude-request*` opens. She writes her question there, in as many lines as she likes, for instance:

```
Write a naive Fibonacci with declare for speed.
```

and presses `C-c r` again to send it. The answer replaces the question window.

Inserting the code. When an answer contains code, Claude puts it between lines of three back quotes:

```
```lisp
(defun fib (n)
 (declare (type fixnum n))
 (if (< n 2) n (+ (fib (- n 1)) (fib (- n 2)))))
```
```

Each such block is a snippet. The `C-c y` command inserts all the snippets of the last answer at the cursor of Nia's file, after asking, and closes the answer window. It works from the file window and from the answer window. If Nia does not want the code, `C-x 1` closes the answer window.

Discussion. For a longer conversation, Nia presses `C-c r` and then `d`. The `*discussion*` window stays open, whatever Nia does in the other windows. She writes after the last `You:` line and presses `C-c r`; Claude's answer is appended, followed by a new `You:.` Each message sends the whole discussion and the file around its cursor, so Claude sees the code as it is now. To bring a snippet of the discussion into the file, Nia puts the cursor inside the snippet and presses `C-c t`, for tangle; SBEmacs asks, then inserts the snippet at the cursor of the file. The `C-x 0` command closes the discussion window, and `C-c r d` brings it back as it was.

Codex and ChatGPT. Codex has the same distinction between a short question and a continuing discussion. Nia presses `C-c g` and sees this menu:

```
h  Hints about the code at the cursor
q  Question in a window of its own; C-c g sends it
d  Discussion that stays open; C-c g sends each turn
s  Set-up: is Codex installed and signed in with ChatGPT?
```

For a question, she chooses `q`, writes as many lines as she needs, and presses `C-c g`. The answer replaces the question window. If it contains fenced snippets, `C-c y` asks for confirmation, inserts all of them at the cursor in the main file, and closes the help window.

For a conversation, she chooses `d`. She writes after the last `You:` and presses `C-c g` for every turn. Codex receives the discussion so far and a fresh view of the source around the cursor. The window deliberately remains open. To use one answer, she places the cursor inside that snippet and presses `C-c t`. She closes the discussion herself with `C-x 0` when she is satisfied; `C-c g d` brings the same discussion back.

Both forms are read-only until Nia explicitly accepts text. Codex runs in a read-only sandbox and an ephemeral command-line session. It cannot edit the project, and the instructions tell it to avoid the editor's C sources. The implementation is the Common Lisp script `lisp/extensions/codex.lisp`; no C code is needed for this feature.

Help page. The help page of `C-h` has the same keys: from there, `C-c r` asks Claude, and `C-c g` asks ChatGPT. Any other key takes Nia back to her file and closes the windows of the assistants, except the discussion.

2.10 Executing Lisp programs

There are many ways of executing Lisp in SBEmacs. They all evaluate the expressions in the same Common Lisp, the one that runs the editor. Errors are reported on the message line; they never take the editor down.

The Lisp prompt. As you learned in Chapter 1, `Esc :` prompts for an expression and shows its value.

Evaluating a definition. Nia writes a program in a file, for instance `fib.lisp`:

```
(defun fib (n)
  (if (< n 2)
      n
      (+ (fib (- n 1)) (fib (- n 2))))))
```

With the cursor after the last parenthesis, she presses `C-x C-e`. SBEmacs evaluates the expression before the cursor and shows `FIB` on the message line, which means that the function is defined. If the cursor is anywhere inside the definition, `C-M-x` has the same effect. Now she can call the function from the Lisp prompt:

```
Esc :
> (fib 20) Enter
6765
```

She could also type `(fib 20)` in the file and press `C-x C-e` after it. The `Esc ]` command evaluates the expression around the cursor and inserts the value into the text.

The editor from Lisp. The expressions can control the editor. For instance, `(insert "Hello")` inserts a text at the cursor, `(goto-line 10)` moves the cursor, `(buffer-filename)` returns the name of the file.

Running a program in SBCL. A program that reads files or prints a lot is better run by SBCL itself, outside the editor. Nia adds a line that prints a result at the end of `fib.lisp`,

```
(format t "fib(25) = ~a~%" (fib 25))
```

saves the file, and presses `Esc !` (or `C-x @`) to run a shell command:

```
sbcl --script fib.lisp
```

SBEmacs shows the output in a buffer of its own:

```
fib(25) = 75025
```

Configuration. When SBEmacs starts, it loads `~/.sbemacs/init.lisp`, if the file exists. That is where Nia puts her own commands and keybindings:

```
(defun insert-date ()
  (multiple-value-bind (s m h d mo y) (get-decoded-time)
    (declare (ignore s m h))
    (insert (format nil "~D-~2,'0D-~2,'0D" y mo d))))

(global-set-key "C-c d" 'insert-date)
(set-color :comment 244) ; grey comments
```

Keys are written as in GNU Emacs: `"C-c d"`, `"M-o"`, `"C-M-y"`, `"F1"`.

Scripts. Syntax colouring, indentation, keybindings and the assistants are Lisp scripts, in the `lisp/` folder of the sources. The files that Nia puts in `~/.sbemacs/extensions/` and `~/.sbemacs/languages/` are loaded too. If she edits any of these files, she presses `C-x C-r`, and SBEmacs reloads what changed; there is no need to run `make` again. For instance, a new language is a file with two forms:

```
(define-language "Go"
  :extensions '(".go")
  :line-comment "//"
  :block-comment '("/*" . "*/")
  :keywords '("func" "package" "import" "return" "if" "else"))

(define-indentation "Go" :style :c :width 4)
```

Chapter 4 explains how to ask Claude Code to write such scripts for you.

2.11 Editing and running Nim

Nim mode lets Nia edit, compile and run a program without leaving SBEmacs. It provides syntax coloring, two-space indentation, comments, and navigation between procedures, functions, iterators, templates and macros. The terminal and GUI versions use the same Lisp callbacks.

Installation and loading. Install Nim and make sure its `nim` command is on the path seen by SBEmacs. Press `M-!`, type `nim --version` and press `C-c s` or click SEND. The lower window displays the compiler version. `C-x 1` returns to the file. If macOS blocks a downloaded Nim executable, approve the trusted compiler with Open Anyway in System Settings, Privacy & Security.

Build and install the updated SBEmacs:

```
make
sudo make install
```

Restart the editor. Files ending in `.nim`, `.nims` and `.nimble` automatically use Nim mode. Its implementation is `lisp/modes/nim-mode.lisp`. To reload it, use `M-x nim-mode`. Alternatively, run `M-x load-mode`, enter `nim` at the Load mode prompt, and press Enter. TAB completes names. Mode lookup starts in the editor's modes directory, independently of the shell directory. A personal copy in `~/.sbemacs/modes/nim-mode.lisp` takes precedence when explicitly loading the mode.

Writing a program. Press `C-x C-f` and enter `~/wrk/hi.nim`. The folder must already exist. Write:

```
import os

echo "Hello, ", paramStr(1)
```

SAVE or **C-x C-s** saves the source. TAB indents the current line; repeated TAB cycles alternative levels. Enter indents the new line. The **#** button or **M-;** comments or uncomments a line or selected region. **C-x C-i** indents a selected region.

Coloring recognizes nested comments, documentation comments, raw and triple-quoted strings, backtick identifiers, and numeric suffixes. It handles unfinished code while Nia types. Indentation is an editing aid, not a complete Nim parser. Macros can introduce syntax beyond its rules; use TAB to choose another level when necessary.

Build and Run. Click MORE and select Build or Run. Use arrows and Enter, or click the option and then RUN. The named commands are:

| | |
|------------------------------------|-----------------------------|
| M-x nim-build | save and compile |
| M-x nim-run | save, compile and run |
| M-x nim-next-definition | next routine definition |
| M-x nim-previous-definition | previous routine definition |

Build and Run work on `.nim` files; coloring also covers `.nims` and `.nimble`. Both commands save the current source before compiling and stop if saving fails. Build calls `nim c`. Run calls `nim c -r` and reads arguments in the minibuffer. For `hi.nim`, enter Edward at Nim program arguments:. The output window shows:

```
$ nim c -r --hints:off 'hi.nim' Edward
Hello, Edward
```

The command runs in the source file's directory. Compiler diagnostics and program output appear below the source; a failed command also shows its exit status. **C-x 1** closes the output window and returns to the source. Edit and use Run again to test the changed program.

Using the shell. For additional compiler options, press **M-!** or **C-x @** from the source window and write:

```
nim c -r --hints:off --nimcache:xx hi.nim Edward
```

SEND or **C-c s** executes the command. Each submission starts a new shell: `cd` does not change the directory of the next submission. On macOS and Linux, put dependent commands together:

```
cd "$HOME/wrk" && nim c -r hi.nim Edward
```

Entered lines are joined as command continuations, not separate shell commands. Windows uses `cmd.exe` and its own quoting conventions.

Execution is synchronous: SBEmacs waits for the command to finish. Its shell capture supplies empty standard input and cannot accept interactive keystrokes or passwords. Supply arguments, a file, or a pipe. For example, on macOS or Linux:

```
printf '3 4 +\n' | ./rdwrt
```

Arguments entered at the Nim prompt are shell text; quote filenames containing spaces. The output buffer does not interpret terminal colors or cursor-control sequences.

The Nim tutorial. The nimacros repository supplies the mode and installation instructions in `sbemacs/`. Hello World, State Machines and Shell are the first three chapters in `nimacros/chaps/nimdoc1.md`. They now use SBEmacs for editing and execution while retaining the author’s explanations. The existing tutorial PDF has not been regenerated. These local changes must be published before GitHub shows the updated instructions:

<https://github.com/FemtoEmacs/nimacros>

The compiler tests ran Hello World, Zeller’s congruence, and both Fibonacci implementations. Zeller, run inside SBEmacs with arguments `2016 8 31`, displayed `3`, the number for Wednesday.

Chapter 3

Shell

Nia, a Greek young woman, has an account on a well known social network. She visits her friends' postings on a daily basis, and when she finds an interesting picture or video, she presses the *Like*-button. However, when she needs to discuss her upcoming holidays on the Saba Island with her Argentinian boyfriend, she uses the live chat box. After all, hitting buttons and icons offers only a very limited interaction tool, and does not produce a highly detailed level of information that permits the answering of questions and making of statements.

Using a chat service needs to be very easy and fun, otherwise all those teenage friends of Nia's would be doing something else. I am telling you this, because there are two ways of commanding a computer. The first is called Graphical User Interface (GUI) and consists of moving the cursor with a mouse or other pointing device and clicking over a menu option or an icon, such as the *Like* button. As previously mentioned, a Graphical User Interface often does not generate adequate information for making a request to the computer. In addition, finding the right object to press can become difficult in a labyrinth of menu options.

The other method of interacting with the computer is known as Shell and is similar to a chat with one's own machine.

In your computer, there is a giant program, called the operating system, which controls all peripherals that the machine uses to stay connected with the external world: pointing devices, keyboard, video terminal, robots, cameras, solid state drives, pen drives, and other peripherals.

In a Shell interface, Nia issues written instructions that the operating system answers by fulfilling the assigned tasks. The language that Nia uses to chat with the operating system is called *Bourne-again shell*, or *bash* for short. This language has commands to go through folders, browser files, create new directories, copy objects from one place to the other, configure the machine, install applications, change the permissions of a file, create groups to organize users and devices, etc. Recent Macintosh computers use *zsh*, a cousin of *bash*; for the commands of this

chapter, the two languages are the same. When accessing the operating system through a text-based terminal, a shell language is the main way of executing programs and doing work on a computer.

The shell interface derives its name from the fact that it acts like a shell surrounding all other programs being run, and controlling everything the machine performs.

To make a long story short, it is much faster to complete tasks using a shell than with graphical applications, icons, menus and mouse. Another benefit of the shell is that you can gain access to many more commands and scripts than with a Graphical User Interface.

In order not to scare off the feeble-minded, many operating systems hide access to the text terminal. In some distribution of Linux, you need to maintain the **[Alt]** key down, then press the **[F2]** key to open a dialog box, where you must type the name of the terminal you want to open. If you are really lucky, you may find the icon of the terminal on the toolbar.



If the way of opening the text terminal is not obvious, you should ask for help from a student majoring in Computer Science. You can teach her Russian, Sanskrit, Javanese or Ancient Greek, as a compensation for the time that she will spend explaining how to start a terminal in OS X or Linux. If you are afraid of paying for a few minutes of tutorial about the Bourne-again shell with hundred hours of classes on Russian, don't worry! The CS major will give up after barely starting the section on the alphabet. After all, Philology is much more difficult than Computer Engineering. For details, read the tale "The man who could speak Javanese" by Lima Barreto.

The prompt. The shell prompt is where one types commands. The prompt has different aspects, depending on the configuration of the terminal. In Nia's machine, it looks something like this:

```
~$ _
```

Files are stored in folders. Typically, the prompt shows the folder where the user is currently working.

The main duty of the operating system is to maintain the contents of the mass storage devices in a tree structure of files and folders. Folders are also called *directories*, and like physical folders or cabinets, they organize files.

A folder can be put inside another folder. In a given machine, there is a folder reserved for duties carried out by the administrator. This special folder is called home or personal directory.

Besides the administrator, a machine can have other rightful users, each with a personal folder. For instance, Nia's folder on her Mac OS X has the `/Users/nia` path. You will learn a more formal definition of path later on.

pwd # The `pwd` command informs the user's position in the file tree. A folder can be placed inside another folder. For example, in a Macintosh, Nia's home folder is inside the `/Users` directory.

One uses a path to identify a nest of folders. In a path, a subfolder is separated from the parent folder by a slash bar. If one needs to know the current folder, there is the `pwd` command.

```
~$ pwd          # shows the current folder. 
/Users/nia
```

When Nia issues a command, she may add comments to it, so her boyfriend that does not know Bourne-again shell (bash) can understand what is going on and learn something in the process. Comments are prefixed with the `#` hash char, as you can see in the above chat. Therefore, when the computer sees a `#` hash char, it ignores everything to the end of the line.

mkdir wrk # creates a `wrk` folder inside the current directory, where `wrk` can be replaced with any other name. For instance, if Nia is inside her home directory, `mkdir wrk` creates a folder with the `/Users/nia/wrk` path.

cd wrk # One can use the `cd <folder name>` commands to enter the named folder. The `cd ..` command takes Nia to the parent of the current directory. Thanks to the `cd` command, one can navigate through the tree of folders and directories.

Tab. If you want to go to a given directory, type part of the directory path, and then press . The shell will complete the folder name for you.

Home directory. One can use a `~` tilde to represent the home directory. For instance, `cd ~` will send Nia to her personal folder. The `cd $HOME` has the same effect, i.e., it places Nia inside her personal directory.

echo # One can use the `echo` command to print something. Therefore, `echo $HOME` prints the contents of the `HOME` environment variable on the terminal.

Environment variables store the terminal configuration. For instance, the `HOME` variable contains the user's personal directory identifier. One needs to prefix the environment variable with the `$` char to access its contents. Therefore, `echo $HOME` displays the contents of the `HOME` variable.

The instruction `echo "Work Space" > readme.txt` creates a `readme.txt` text file and writes the "Work Space" string there. If the `readme.txt` file exists, this command supersedes it.

The command `echo "Shell practice" >> readme.txt` appends a string to a text file. It does not erase the previous content of the `readme.txt` file. Of course, you should replace the string or the file name, as necessity dictates.

Below you will find an extended example of a chat between Nia and the OS X operating system.

```
~$ mkdir wrk      # creates a work space folder      Enter
~$ cd wrk        # transfers action to the wrk file   Enter
~/wrk$ echo "* Work space" > readme.txt               Enter
~/wrk$ echo "This folder is used" >> readme.txt        Enter
~/wrk$ echo "to practice the bash" >> readme.txt       Enter
~/wrk$ echo "commands and queries." >> readme.txt      Enter
~/wrk$ ls        # list files in current folder.      Enter
readme.txt
~/wrk$ cat readme.txt # shows the file contents.      Enter

* Work space
This folder is used
to practice the bash
commands and queries.
```

ls # By convention, a file name has two parts, the id and the extension. The id is separated from the extension by a dot. The `ls` command lists all files and subfolders present in the current folder. The `ls *.txt` prints only files with the `.txt` extension.

Wild card. The `*.txt` pattern is called wild card. In a wild card, the `*` asterix matches any sequence of chars, while the `?` interrogation mark matches a single char.

ls -lia *.txt # prints detailed information about the `.txt` files, like date of creation, size, etc.

```
~/wrk$ ls -la 
```

```
total 8
drwxr-xr-x    3 edu500ac  staff    102 Sep 18 16:29 .
drwxr-xr-x+ 321 edu500ac  staff  10914 Sep 18 14:40 ..
-rw-r--r--    1 edu500ac  staff     74 Sep 18 16:30 readme.txt
```

Files starting with a dot are called hidden files, due to the fact that the `ls` command does not normally show them. All the same, the `ls -a` option includes the hidden files in the listing.

In the preceding examples, the first character in each list entry is either a dash (-) or the letter d. A dash (-) indicates that the file is a regular file. The letter d indicates that the entry is a folder. A special file type that might appear in a `ls -la` command is the symlink. It begins with a lowercase l, and points to another location in the file system. Directly after the file classification comes the permissions, represented by the following letters:

- r – read permission.
- w – write permission.
- x – execute permission.

cp readme.txt lire.txt # makes a copy of a file. You can copy a whole directory with the `-rf` options, as shown below.

```
~/wrk$ ls
readme.txt
~/wrk$ cp readme.txt lire.fr
~/wrk$ ls
lire.fr  readme.txt
~/wrk$ cd ..
~$ cp -rf wrk wsp
~$ cd wsp/
~/wsp$ ls
lire.fr  readme.txt
```

rm lire.txt # removes a file. The `-rf` option removes a whole folder.

```
~/wsp$ ls
lire.fr  readme.txt
~/wsp$ rm lire.fr
~/wsp$ ls
readme.txt
~/wsp$ cd ..
~$ ls wrk
lire.fr  readme.txt
~$ rm -rf wrk
~$ ls wrk
ls: wrk: No such file or directory
```

mv **wsp wrp** # changes the name of a file or folder, or even permits the moving of a file or folder to another location.

```
~$ mv wsp wrk
~$ cd wrk
~/wrk$ ls
readme.txt
~/wrk$ mv readme.txt read.me
~/wrk$ ls
read.me
~/wrk$ cp read.me wrk-readme.txt
~/wrk$ ls
read.me  wrk-readme.txt
~/wrk$ mv wrk-readme.txt ~/Documents/
~/wrk$ ls
read.me
~/wrk$ cat ~/Documents/wrk-readme.txt
Work space
This folder is used
to practice the bash
commands and queries.
```

Copy to pen drive. In most Linux distributions, the pen drive is seen as a folder inside the `/media/nia/` directory, where you should replace **nia** with your user name. In the Macintosh, the pen drive appears at the `/Volumes/` folder. The commands **cp**, **rm** and **ls** see the pen drive as a normal folder.

3.1 Package managers

A package manager is a tool that works with core libraries to handle, even though poorly, the installation and removal of software. Each operating system has its

own package manager. Even if you stick to a single operating system, you will have to deal with many package managers and different philosophies concerning software installation.

I will teach you how to use three installation tools: `apt-get`, `homebrew` and `git clone`. However, this tutorial will not completely cover the many subtleties involved while you make your machine usable. You will really need the help of that girl who is majoring in Computer Engineering. Tell her that, if she teaches you how to configure `apt-get` install, you will teach her the ancient Javanese script.

apt-get. Debian, Ubuntu and their relatives keep the programs in repositories that the `apt-get` tool consults. Before installing, ask `apt-get` to refresh its list of programs; then install what you need:

```
~$ sudo apt-get update
~$ sudo apt-get install sbcl git build-essential libncurses-dev
```

The `sudo` prefix is needed because `apt-get` writes into system folders. Fedora uses `dnf` instead of `apt-get`, with the same syntax.

Homebrew. The Macintosh has no package manager of its own. Homebrew fills the gap. The installation command is on the <https://brew.sh> page; copy it into the terminal and press `[Enter]`. After that, `brew` installs programs without `sudo`:

```
~$ brew install sbcl sdl2 sdl2_ttf pkg-config
~$ brew upgrade          # brings new versions of everything
```

git clone. Many programs, SBEmacs among them, are distributed as source code in a git repository. The `git clone` command makes a copy of the repository, `git pull` brings the changes, and `make` builds the program, as you saw in section 2.1. The advantage is that you always have the most recent version; the drawback is that you must build it yourself.

Where programs go. Unix and Unix like systems are unable to find a command, except when it is installed in a pre-defined search path or when the file path is completely specified. The search path is kept in the `PATH` environment variable:

```
~$ echo $PATH
/opt/homebrew/bin:/usr/local/bin:/usr/bin:/bin
~$ which sbemacs
/usr/local/bin/sbemacs
```

The `which` command tells where a program is. If you installed two copies of a program, the shell runs the one whose folder comes first in `PATH`.

Chapter 4

Teaching Claude: SKILL.md

There are two ways of working with Claude. Inside SBEmacs, `C-c r` asks Claude a question about the text you are editing. Claude only reads: it cannot open files or run programs, and you decide what goes into your buffer. Outside the editor, in the text terminal, Claude Code can do much more. If Nia enters a folder and types `claude`, she can chat with Claude Code, which reads the files of the folder, writes new ones, and runs commands like `make`, always asking her permission first.

Suppose that Nia wants SBEmacs to colour Go programs. She could type `claude` in her terminal and say:

Write an SBEmacs script that colours and indents Go files.

Claude Code knows Common Lisp, but it does not know SBEmacs. It does not know that a language is defined by `define-language`, that personal scripts go to `~/.sbemacs/languages/`, or that `C-x C-r` reloads them. It would find out by reading the sources, which takes time, or it would guess, which is worse. Generative AI are notoriously bad at saying “I don’t know”.

A skill solves this problem. A skill is a folder with a file called `SKILL.md`, which teaches Claude how to do one kind of task. Claude Code reads only the first lines of each skill, its description. When a request matches the description, Claude reads the whole file and follows it.

4.1 The format of SKILL.md

A `SKILL.md` file has two parts. The first part, between two lines of three dashes, is a header with the name of the skill and its description:

```
---
name: sbemacs-scripts
description: Write or change SBEmacs Lisp scripts (key bindings,
```

```

    commands, syntax colouring with define-language, indentation ...)
    Use when the user wants SBEmacs to do something new, or asks
    how its Lisp API works.
---

```

The description is the most important line of the file, because Claude uses it to decide whether the skill applies. It should say what the skill does and when to use it. The `name` becomes a command: Nia can call the skill explicitly by typing `/sbemacs-scripts` in Claude Code.

The second part is ordinary text, in Markdown, the format of `README.md` files: headings start with `#`, lists with `-`, and program examples go between lines of three back quotes. This part contains the instructions: where things are, which rules to follow, and examples to imitate. Write it as you would explain the task to a new colleague, with short sentences and concrete examples.

4.2 Where skills live

Claude Code looks for skills in two places:

- `~/.claude/skills/NAME/SKILL.md` – personal skills, which work in every folder of your machine.
- `.claude/skills/NAME/SKILL.md` – project skills, inside a repository. They work when Claude Code runs in that repository, and everybody who clones the repository gets them.

The SBEmacs repository brings a project skill:

```
~/csand/sbemacs/.claude/skills/sbemacs-scripts/SKILL.md
```

The folder name starts with a dot, so `ls` does not show it; `ls -a` does. If Nia runs `claude` inside `~/csand/sbemacs`, the skill is available. To use it everywhere, for instance in the folder where she keeps her Go programs, she copies it to her personal skills:

```

~$ mkdir -p ~/.claude/skills
~$ cp -r ~/csand/sbemacs/.claude/skills/sbemacs-scripts
↪ ~/.claude/skills/

```

4.3 Using the skill

Nia enters her folder, starts Claude Code and asks:

```
~/go-programs$ claudé
> Write an SBEmacs script that colours and indents Go files.
```

Claude recognizes the task from the description, reads the skill, and writes the file `~/sbemacslanguages/go.lisp`, with a `define-language` form and a `define-indentation` form. It asks before creating the file. Then Nia opens a Go file in SBEmacs, or presses `C-x C-r` if the editor is already running, and the colours appear. If Claude does not pick the skill by itself, Nia names it:

```
> /sbemacslscripts a command on C-c d that inserts today's date
```

4.4 The SBEmacs skill

Here is the complete file. Its rules are the rules of this book: the assistant never changes the text behind your back, errors end in a message, and programs declare their types instead of hiding the warnings of the compiler.

```
---
name: sbemacslscripts
description: Write or change SBEmacs Lisp scripts (key bindings, commands,
  ↳ syntax colouring with define-language, indentation with define-indentation,
  ↳ colours, extensions) for SBEmacs (Steel Bank Emacs), the Emacs-like editor
  ↳ written in C and SBCL. Use when the user wants SBEmacs to do something new,
  ↳ or asks how its Lisp API works.
---

# Writing SBEmacs scripts

SBEmacs (Steel Bank Emacs) is a small editor core in C driven by SBCL.
  ↳ Everything a user
  customises is Common Lisp, in *scripts* that the editor reloads with
  `C-x C-r`: no `make` is needed. Work in scripts; touch `src/*.c` or the
  four engine files (`lisp/package.lisp`, `ffi.lisp`, `loader.lisp`,
  `core.lisp`) only when the user explicitly asks for a change to the core,
  and then remind them to run `make && sudo make install`.

## Where scripts live

Folder	Package	Use it for
`~/sbemacslinit.lisp`	`SBEMACS-USER`	the user's settings and small
↳ commands		
`~/sbemacslextensions/*.lisp`	`SBEMACS-USER`	the user's larger
↳ extensions		
`~/sbemacsllanguages/*.lisp`	`SBEMACS-USER`	the user's languages
`lisp/` in the sources	`SBEMACS`	only when changing SBEmacs itself
```


``SBEMACS-USER`` uses ``COMMON-LISP`` and ``SBEMACS``, so only exported symbols are available there; do not reach into ``sbemacs::`` internals from a user script. Load order: ``highlight``, ``indent``, ``theme``, ``undo``, ``gui``, ``languages/*``, ``defaults``, ``extensions/*`` (among them ``menu``, the mode line's buttons), then the user's languages, extensions and ``init.lisp``.

Rules

1. ****First Law: never change the user's text behind their back.**** A command changes the buffer only when the user runs it, and asks first (``prompt`` for y/n) before anything large or destructive.
2. ****Errors end in a message, never a crash.**** Check inputs; report with ``(message "...)"``. Commands run inside a handler, but do not rely on it.
3. ****Declare, do not muffle.**** Use ``(declaim (ftype (function (...)) ...) name))``, ``(declare (type ...))`` and ``(declaim (optimize (safety 3)))``. Never muffle warnings; fix them.
4. ****`defparameter`, not `defvar`, for values that must change when the script is reloaded.**** ``defvar`` keeps the old value, so an edit to it seems to have no effect after ``C-x C-r``.
5. ****Keys in Emacs notation:**** ``"C-c d"``, ``"M-o"``, ``"C-M-f"``, ``"C-x h"``, ``"F1"``. Every key reaches the Lisp keymap before the C core, so any key can be bound. Leave ``C-c <letter>`` keys to the user unless asked.
6. ****Define commands with `defcommand`**** (same syntax as ``defun``) so that ``M-x`` finds them.
7. ****Positions are byte offsets**** into UTF-8 text; ``buffer-substring`` returns a string, ``buffer-octets`` the bytes.
8. ****Undo is automatic.**** Every change is recorded; all the changes made by one key are undone together by ``C-/``. Do not record anything yourself.
9. Test pure functions with ``make test`` (``tests/run-tests.lisp``, forms ``(check "name" form expected)``).

The API (package SBEMACS)

- Moving: ``point`` ``goto-char`` ``forward-char`` ``backward-char`` ``next-line`` ``previous-line`` ``beginning-of-line`` ``end-of-line`` ``beginning-of-buffer`` ``end-of-buffer`` ``goto-line`` ``line-start`` ``line-end`` ``line-number``
- Reading: ``buffer-size`` ``buffer-substring`` ``buffer-octets`` ``char-after`` ``current-line-text`` ``search-forward`` ``search-backward``
- Changing: ``insert`` ``delete-char`` ``backward-delete-char`` ``kill-region`` ``copy-region`` ``yank`` ``kill-line`` ``undo`` ``redo``
- Mark and region: ``mark`` ``set-mark`` (activates the shaded region) ``push-mark`` (does not) ``region-active-p`` ``deactivate-mark``
- Buffers and windows: ``get-buffer-name`` ``buffer-filename`` ``buffer-modified-p`` ``select-buffer`` ``find-file`` ``save-buffer`` ``kill-buffer`` ``split-window`` ``other-window`` ``delete-window`` ``delete-other-windows`` ``window-count``
- Talking to the user: ``message`` ``clear-message-line`` ``prompt``

```

`update-display` `get-key` (returns the typed text, or "" for a bound
key, then `get-key-name` gives its name) `set-buffer-hint` (text at the
end of a buffer's mode line) `*mode-line-menu*` (the mode line's
clickable buttons, entries `(LABEL KEY-OR-COMMAND)`; the whole mode line
is composed in `lisp/extensions/menu.lisp`, `mode-line-pieces`)
`*mode-line-buttons-function*` (choose the buttons per buffer)
`usage-statistics` (`*command-usage*`, `*button-usage*`)
- Keys and commands: `global-set-key` `global-unset-key` `key-binding`
`defcommand` `execute-builtin` (runs a command of the C core by name)
`run-key-as-typed` (does what typing a key does, Lisp or C binding)
- Looks: `set-color` (faces `:keyword` `:comment` `:block-comment`
`:string` `:digits` `:alpha` `:symbol` `:brace` `:modeline` `:region`,
`:menu` `:menu-pressed` (the mode line's buttons), and for prose
`:heading` `:emphasis` `:strong` `:link`;
colours `:red`..., 0-255 or `"#rrggbb"`)
- Languages: `define-language` (`:extensions` `:line-comment`
`:block-comment` `:strings` `:keywords` `:escape` `:char-prefix`
`:case-insensitive` `:word-chars` `:backslash-commands`, and
`:highlighter` for a language that is not made of tokens: see
`lisp/languages/markdown.lisp`),
`define-indentation` (`:style` `:c` `:lisp` `:python` `:prolog` `:block`
`:markdown`, `:width`, `:tabs`, `:cycle`, `:levels`, `:newline`),
`define-indent-style`
- Shell: `shell-command`

```

Recipes

A command on a key:

```

```lisp
(defcommand insert-date ()
 "Insert today's date at the cursor."
 (multiple-value-bind (s m h d mo y) (get-decoded-time)
 (declare (ignore s m h))
 (insert (format nil "~D-~2,'0D-~2,'0D" y mo d))))

(global-set-key "C-c d" 'insert-date)
```

```

A language (one file in `~/sbemacslanguages/`):

```

```lisp
(define-language "Go"
 :extensions '(".go")
 :line-comment "//"
 :block-comment '("/" . "*/")
 :strings '("\\""")
 :keywords '("func" "package" "import" "return" "if" "else" "for"))

(define-indentation "Go" :style :c :width 4)

```

```

```

Reading keys in a small modal loop (as the help page does):

```
```lisp
(loop
  (message "y: yes, any other key: no")
  (update-display)
  (let ((k (get-key)))
    (return (string= k "y"))))
---
```

After writing a script

Tell the user where the file is and that ``C-x C-r`` (or restarting the editor) loads it; name the key or the ``M-x`` command that uses it.

4.5 Writing your own skill

A skill is worth writing when you find yourself explaining the same thing to Claude more than twice. For instance, if Nia always wants her Lisp programs with type declarations and tests, she can create `~/.claude/skills/nia-lisp/SKILL.md`:

```
---
name: nia-lisp
description: Nia's way of writing Common Lisp. Use for any
  Common Lisp code she asks for.
---

- Declare the types of every function with declaim ftype.
- Never muffle warnings; fix them.
- Write a test for each function with the check macro.
- Explain each function in one sentence, in plain English.
```

Keep a skill short and specific. Claude reads it when it applies, so a long skill costs time on every task of that kind, and a vague description makes Claude use the skill at the wrong moments, or never.

About SBEmacs

SBEmacs, Steel Bank Emacs, is a tiny Emacs clone, originally derived from a minimalist editor called Atto. Although built upon the C Atto codebase, SBEmacs can be customized using Lisp: Steel Bank Common Lisp, or SBCL for short. Its predecessor, FemtoEmacs, used a Lisp of its own, femtolisp.

The fundamental paradigm of Emacs is that it rests on a small core, but can be extended and customized through the Lisp programming language.

Instead of requiring from the programmer a compile, build, test and fix cycle — Lisp offers a Read, Eval, Print Loop (REPL).

1. **Read.** The REPL reads a symbolic expression such as `(+ 42 13 4)` from a text terminal. The symbolic expression — *sexpr* — always has the form: (operation $a_1 a_2 \dots a_n$), where $a_1, a_2 \dots a_n$ are the arguments of the operation.
2. **Eval.** The compiler translates the symbolic expression to machine language, then executes the generated code and calculates the corresponding value.
3. **Print.** The value produced in step 2 is printed.
4. **Loop.** The REPL returns to step 1.

Femtolisp, the Lisp of FemtoEmacs, did not generate native code, but compiled symbolic expressions to a virtual machine language. SBCL, on the other hand, compiles every expression to the machine language of your computer, so the scripts that colour and indent your programs run as fast as the C core of the editor.

The REPL approach enables fast prototyping of extensions as each idea can be tested individually on the command line inside the editor with an immediate result check. In SBEmacs, this is true of the editor itself: syntax colouring, indentation, the key bindings and the assistants are Lisp scripts. If you edit one of them and press `C-x C-r`, the editor reloads it; there is no need to build the program again.

For historical reasons, the reference GNU Emacs implementation uses a specialised version of lisp known as Emacs Lisp. However, the Emacs Lisp dialect is not standard lisp, being used only by the GNU Emacs editor and its derivatives.

Common Lisp was chosen as the extension language for SBEmacs for the following reasons:

1. SBCL is open source.
2. Common Lisp is an ANSI standard. One can use classical Common Lisp tutorials and books to learn how to configure and customize SBEmacs.
3. SBCL is fast.
4. SBCL runs on Linux, Macintosh and Windows.

The present text is based on *An introduction to femtolisp*, by Eduardo Costa and Vernon Sipple.

SBEmacs was written by Eduardo Costa, on top of FemtoEmacs by Hugh Barney and its contributors.